

Обо мне

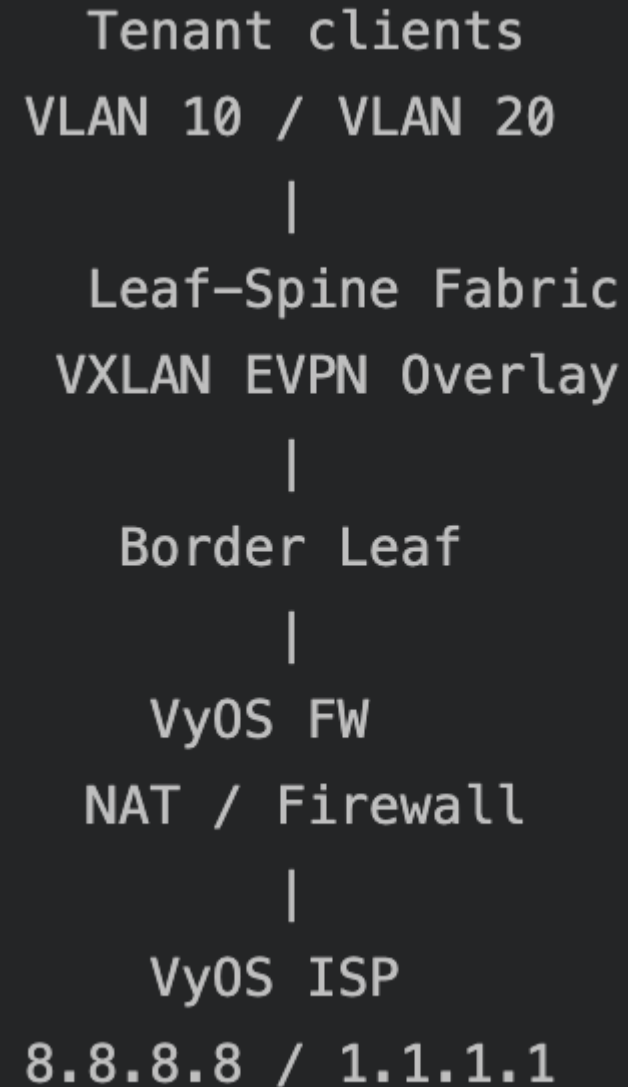
Дипломный проект OTUS: сеть датацентра на VXLAN EVPN

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- СЕТЕВОЙ ИНЖЕНЕР & АРХИТЕКТОР
в Space307
- Курс: сети ЦОД / OTUS
- Стенд собран в EVE-NG
- Оборудование: Arista vEOS, VyOS, VPCS
- Фокус проекта: underlay, overlay, edge,
NAT, multihoming и DCI



Цель проекта

- Построить leaf-spine сеть датацентра на VXLAN EVPN
- Разделить транспортный underlay и клиентский overlay
- Реализовать L2VNI, L3VNI и Anycast Gateway
- Добавить внешний выход через border leaf и VyOS
- Проверить NAT, фильтрацию, отказоустойчивые подключения клиентов и DCI



Использованная архитектура

- DC1: eBGP IPv4 underlay + eBGP EVPN overlay
- DC2: OSPF underlay + iBGP EVPN overlay
- DC2 spine: route-reflector для EVPN
- DCI: eBGP IPv4 underlay + eBGP EVPN overlay между border leaf
- Edge: VyOS BGP, NAT, firewall, route-map и prefix-list

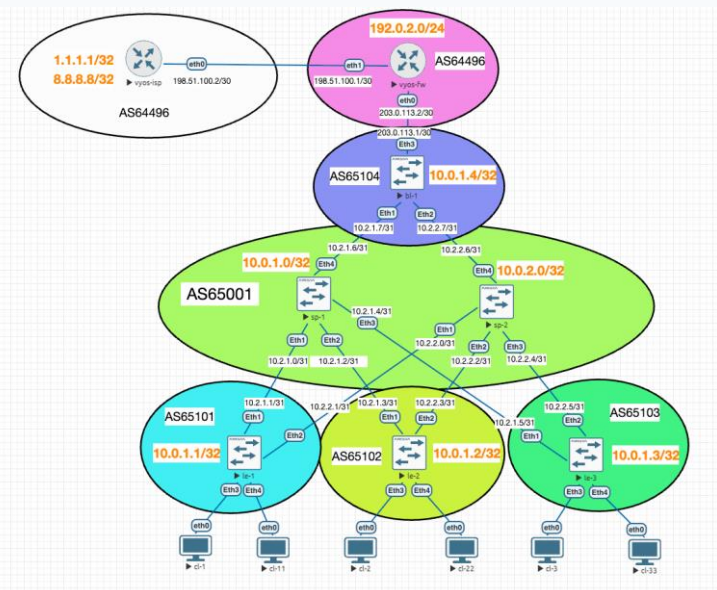
```
sp-1#show bgp evpn summary
BGP summary information for VRF default
Router identifier 10.0.1.0, local AS number 65001
Neighbor Status Codes: m - Under maintenance
Neighbor V AS      MsgRcvd  MsgSent  InQ  OutQ  Up/Down  State  PfxRcd  PfxAcc
10.0.1.1 4 65101    10843    10912    0    0  07:20:51 Estab   4       4
10.0.1.2 4 65102    107948   108247    0    0   3d04h Estab   4       4
10.0.1.3 4 65103    108210   108205    0    0   3d04h Estab  13      13
10.0.1.4 4 65104     13723    13710    0    0  09:09:05 Estab  25      25
10.0.1.5 4 65105    108281   108195    0    0   3d04h Estab  10      10
sp-1#
```

```
sp-21#
sp-21#
sp-21#show ip ospf neighbor
Neighbor ID      Instance VRF      Pri State          Dead Time   Address      Interface
10.0.21.2        1       default 0    FULL            00:00:36   10.22.1.3    Ethernet2
10.0.21.1        1       default 0    FULL            00:00:32   10.22.1.1    Ethernet1
10.0.21.3        1       default 0    FULL            00:00:31   10.22.1.5    Ethernet3
10.0.21.4        1       default 0    FULL            00:00:32   10.22.1.7    Ethernet4
sp-21#
```

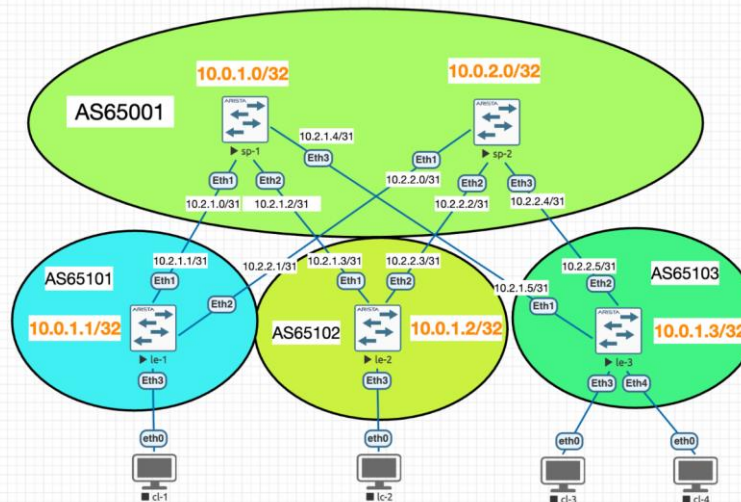
Эволюция стенда

- Стартовая база — lab-06
- Первый этап — один DC с VXLAN EVPN
- Далее — border leaf, VyOS edge, NAT и внешний BGP
- Следующий этап — MLAG, EVPN multihoming, IDRAC VRF
- Финальный этап — DC2/POD и DC1

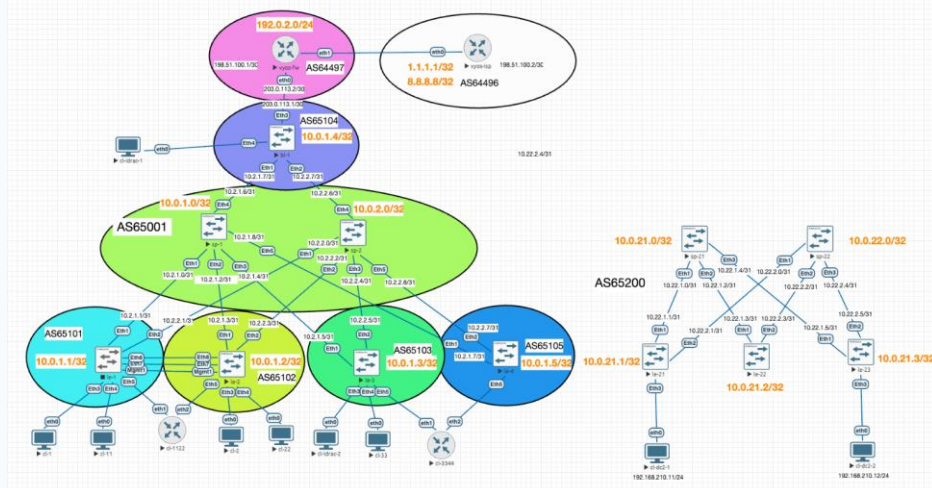
eve-ng-scheme-mvp.png



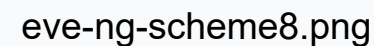
eve-ng-scheme.png



eve-ng-scheme6.png



- DC1: 2 spine, 4 leaf, 1 border leaf
- DC2: 2 spine, 3 leaf, 1 border leaf
- DCI между bl-1 и bl-21
- Внешний контур: vyos-fw, vyos-fw-dc2
vyos-isp
- Клиенты: access, MLAG, EVPN
multihoming, IDRAC, DCI VLAN 10



Адресный план и роли

- DC1 spine: AS 65001; DC1 leaf: AS 65101–65105
- DC2: AS 65200; VyOS: AS 64497/64498; ISP: AS 64496
- VLAN 10/20 → TENANT-1; VLAN 30 → IDRAC; VLAN 210 → TENANT-DC2
- L2VNI: 10010, 10020, 10030, 10210
- L3VNI: 50000 для TENANT-1, 50200 для TENANT-DC2

ASN

Устройство	ASN
sp-1, sp-2	65001
sp-21, sp-22, le-21, le-22, le-23, bl-21	65200
le-1	65101
le-2	65102
le-3	65103
bl-1	65104
le-4	65105
vyos-fw	64497
vyos-fw-dc2	64498
vyos-isp	64496

Tenant-сети

VLAN	L2VNI	Route Target	Anycast Gateway
10	10010	10:10010	192.168.10.1/24
20	10020	20:10020	192.168.20.1/24
30	10030	30:10030	198.51.100.129/25 на bl-1
210	10210	210:10210	192.168.210.1/24

DC1 Underlay

- Underlay DC1 построен на eBGP IPv4
- Все spine-leaf линки — L3 p2p /31
- В BGP underlay анонсируются только loopback /32
- P2P-сети не попадают в BGP RIB через redistribute connected
- Loopback0 используется как VTEP и source для overlay EVPN

```
sp-1#show bgp su
BGP summary information for VRF default
Router identifier 10.0.1.0, local AS number 65001
Neighbor      AS Session State AFI/SAFI           AFI/SAFI State  NLRI Rcd  NLRI Acc
-----
10.0.1.1      65101 Established L2VPN EVPN         Negotiated      4        4
10.0.1.2      65102 Established L2VPN EVPN         Negotiated      4        4
10.0.1.3      65103 Established L2VPN EVPN         Negotiated     13       13
10.0.1.4      65104 Established L2VPN EVPN         Negotiated     25       25
10.0.1.5      65105 Established L2VPN EVPN         Negotiated     10       10
10.2.1.1      65101 Established IPv4 Unicast       Negotiated      1        1
10.2.1.3      65102 Established IPv4 Unicast       Negotiated      1        1
10.2.1.5      65103 Established IPv4 Unicast       Negotiated      1        1
10.2.1.7      65104 Established IPv4 Unicast       Negotiated      6        6
10.2.1.9      65105 Established IPv4 Unicast       Negotiated      1        1
sp-1#
sp-1#
```

DC1 Overlay VXLAN EVPN

- Overlay control-plane — MP-BGP EVPN
- EVPN-соседства строятся по Loopback0
- L2VNI растягивают VLAN поверх fabric
- L3VNI 50000 обеспечивает routing внутри TENANT-1
- Anycast MAC DC1: 00:00:be:ef:ca:fe

```
le-1#show bgp evpn su
BGP summary information for VRF default
Router identifier 10.0.1.1, local AS number 65101
Neighbor Status Codes: m - Under maintenance
Neighbor V AS      MsgRcvd  MsgSent  InQ  OutQ  Up/Down  State  PfxRcd  PfxAcc
10.0.1.0 4 65001    13217   13200    0    0 07:39:10 Estab  52     52
10.0.2.0 4 65001    13261   13144    0    0 07:39:28 Estab  52     52
le-1#show ip route vrf TENANT-1

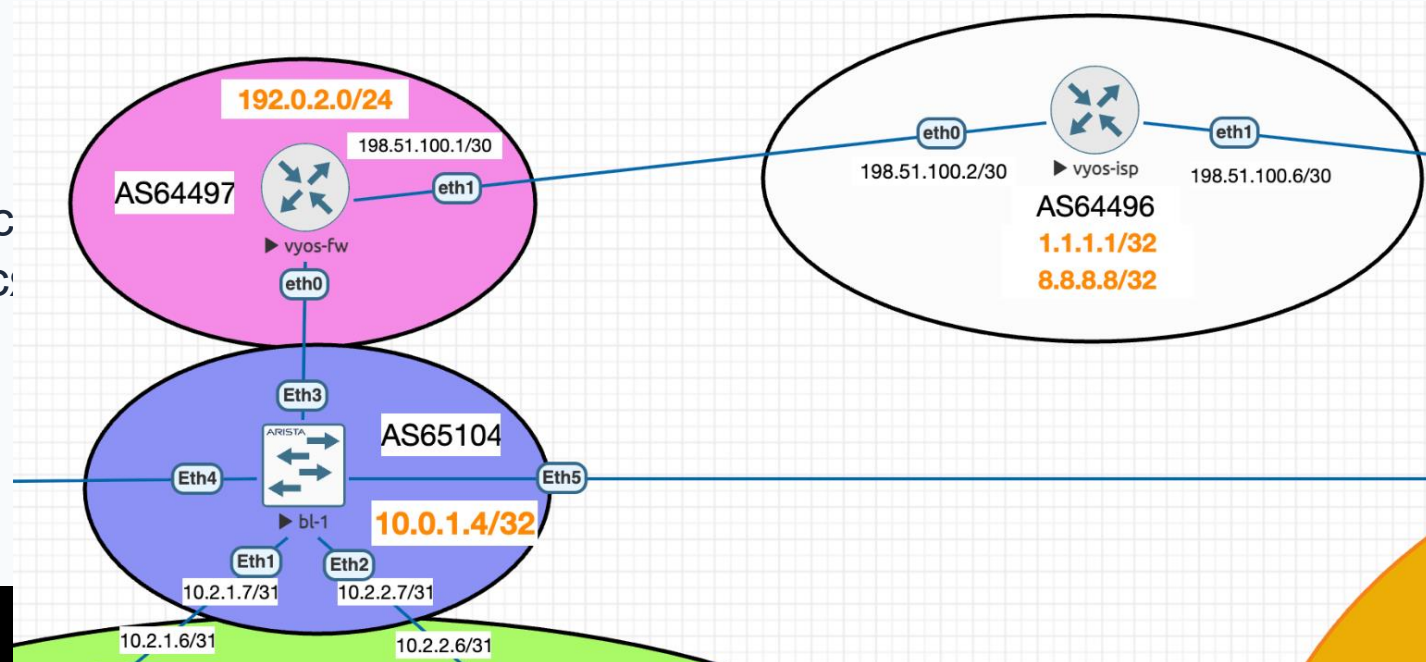
VRF: TENANT-1
Codes: C - connected, S - static, K - kernel,
       0 - OSPF, IA - OSPF inter area, E1 - OSPF external type 1,
       E2 - OSPF external type 2, N1 - OSPF NSSA external type 1,
       N2 - OSPF NSSA external type2, B - Other BGP Routes,
       B I - iBGP, B E - eBGP, R - RIP, I L1 - IS-IS level 1,
       I L2 - IS-IS level 2, O3 - OSPFv3, A B - BGP Aggregate,
       A O - OSPF Summary, NG - Nexthop Group Static Route,
       V - VXLAN Control Service, M - Martian,
       DH - DHCP client installed default route,
       DP - Dynamic Policy Route, L - VRF Leaked,
       G - gRIBI, RC - Route Cache Route

Gateway of last resort:
B E      0.0.0.0/0 [200/0] via VTEP 10.0.1.4 VNI 50000 router-mac 50:01:00:02:f6:c5 local-interface Vxlan1

C        192.168.10.0/24 is directly connected, Vlan10
B E      192.168.20.13/32 [200/0] via VTEP 10.0.1.3 VNI 50000 router-mac 50:01:00:ca:7a:8d local-interface Vxlan1
B E      192.168.20.44/32 [200/0] via VTEP 10.0.1.5 VNI 50000 router-mac 50:01:00:80:9b:62 local-interface Vxlan1
                               via VTEP 10.0.1.3 VNI 50000 router-mac 50:01:00:ca:7a:8d local-interface Vxlan1
C        192.168.20.0/24 is directly connected, Vlan20
B E      203.0.113.0/30 [200/0] via VTEP 10.0.1.4 VNI 50000 router-mac 50:01:00:02:f6:c5 local-interface Vxlan1
```

Border Leaf и внешний выход DC1

- bl-1 — border leaf DC1
- Transit VLAN 100: TENANT-1 → vyos-fw
- Transit VLAN 130: IDRAC → vyos-fw
- vyos-fw отдаёт default route внутрь fabric
- Внутренние tenant-сети не анонсируютснаружу напрямую



```
bl-1#show ip bgp summary vrf TENANT-1
BGP summary information for VRF TENANT-1
Router identifier 203.0.113.1, local AS number 65104
Neighbor Status Codes: m - Under maintenance
```

Neighbor	V	AS	MsgRcvd	MsgSent	InQ	OutQ	Up/Down	State	PfxRcd	PfxAcc
203.0.113.2	4	64497	11385	13653	0	0	09:29:08	Estab	1	1

```
bl-1#
```

```
bl-1#show ip bgp summary vrf IDRAC
BGP summary information for VRF IDRAC
Router identifier 203.0.113.5, local AS number 65104
Neighbor Status Codes: m - Under maintenance
```

Neighbor	V	AS	MsgRcvd	MsgSent	InQ	OutQ	Up/Down	State	PfxRcd	PfxAcc
203.0.113.6	4	64497	8240	9670	0	0	06:51:52	Estab	1	1

```
bl-1#
```

NAT, фильтрация и ISP

- vyos-isp имитирует внешний интернет
- Loopback: 8.8.8.8/32 и 1.1.1.1/32
- На vyos-isp подняты HTTP и iperf3 серверы
- NAT: VLAN 10 → 192.0.2.10; VLAN 20 → 192.0.2.20
- Firewall и route-map ограничивают

```
vyos@vyos-fw:~$ show nat source translations
Pre-NAT      Post-NAT      Proto  Timeout  Mark  Zone
-----
198.51.100.130 198.51.100.130 icmp    26       0
192.168.10.44 192.0.2.10    icmp    8        0
198.51.100.130 198.51.100.130 icmp    28       0
192.168.20.13 192.0.2.20    icmp    18       0
192.168.20.13 192.0.2.20    icmp    25       0
198.51.100.130 198.51.100.130 icmp    23       0
192.168.20.13 192.0.2.20    icmp    10       0
192.168.20.13 192.0.2.20    icmp    13       0
192.168.10.44 192.0.2.10    icmp    14       0
192.168.10.44 192.0.2.10    icmp    1        0
192.168.10.44 192.0.2.10    icmp    25       0
192.168.20.13 192.0.2.20    icmp    14       0
192.168.10.44 192.0.2.10    icmp    16       0
192.168.10.44 192.0.2.10    icmp    17       0
198.51.100.130 198.51.100.130 icmp    4        0
192.168.10.44 192.0.2.10    icmp    21       0
```

```
vyos@vyos-fw:~$ show ip bgp summary
```

```
IPv4 Unicast Summary (VRF default):
BGP router identifier 192.0.2.20, local AS number 64497 vrf-id 0
BGP table version 11
RIB entries 7, using 672 bytes of memory
Peers 1, using 20 KiB of memory
```

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd	PfxSnt	Desc
198.51.100.2	4	64496	4563	4563	11	0	0	10:20:11	3	2	vyos-isp

```
Total number of neighbors 1
```

```
vyos@vyos-fw:~$ show ip bgp
```

```
BGP table version is 11, local router ID is 192.0.2.20, vrf id 0
Default local pref 100, local AS 64497
```

```
Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
               i internal, r RIB-failure, S Stale, R Removed
```

```
Next hop codes: @NNN nexthop's vrf id, < announce-nh-self
```

```
Origin codes: i - IGP, e - EGP, ? - incomplete
```

```
RPKI validation codes: V valid, I invalid, N Not found
```

Network	Next Hop	Metric	LocPrf	Weight	Path
*> 0.0.0.0/0	198.51.100.2	0		0	64496 i
*> 1.1.1.1/32	198.51.100.2	0		0	64496 i
*> 8.8.8.8/32	198.51.100.2	0		0	64496 i
*> 192.0.2.0/24	0.0.0.0	0		32768	i
*> 198.51.100.128/25	0.0.0.0	0		32768	i

Отказоустойчивость клиентов

- cl-1122 подключен через MLAG к le-1/le-2
- cl-3344 подключен через EVPN multihoming к le-3/le-4
- На клиентах используется LACP bonding 802.3ad
- EVPN multihoming передаёт ethernet segment через EVPN
- Link tracking связывает состояние uplink и client-link

```
le-1 (telnet)
le-1#show mlag interfaces

```

mlag	desc	state	local	remote	local/remote status
55	cl-1122	active-full	Po55	Po55	up/up

```
le-1#show mlag
MLAG Configuration:
domain-id          : LEAVES-1-2
local-interface    : Vlan4094
peer-address       : 172.16.101.2
peer-link          : Port-Channel78
hb-peer-address    : 192.168.0.2
hb-peer-vrf        : MGMT
peer-config        : consistent

MLAG Status:
state              : Active

le-3 (telnet)
le-3#show bgp evpn route-type ethernet-segment
BGP routing table information for VRF default
Router identifier 10.0.1.3, local AS number 65103
Route status codes: * - valid, > - active, S - Stale, E - ECMP head, e - ECMP
                   c - Contributing to ECMP, % - Pending BGP convergence
Origin codes: i - IGP, e - EGP, ? - incomplete
AS Path Attributes: Or-ID - Originator ID, C-LST - Cluster List, LL Nexthop - Link Local Nexthop


```

	Network	Next Hop	Metric	LocPref	Weight	Path
* >	RD: 10.0.1.3:1 ethernet-segment	0000:0000:0000:0000:0001	10.0.1.3			
		-	-	0		i
* >Ec	RD: 10.0.1.5:1 ethernet-segment	0000:0000:0000:0000:0001	10.0.1.5			
		10.0.1.5	-	100	0	65001 65105 i
* ec	RD: 10.0.1.5:1 ethernet-segment	0000:0000:0000:0000:0001	10.0.1.5			
		10.0.1.5	-	100	0	65001 65105 i

```
le-3#
le-3#
```

VRF IDRAC

- IDRAC вынесен в отдельный VRF
- VLAN 30 растянут через L2VNI 10030
- Gateway: 198.51.100.129/25 на bl-1
- Public prefix: 198.51.100.128/25
- Доступ из tenant1 демонстрируется через внешний контур

```
vyos@cl-3344:~$ traceroute 198.51.100.143
traceroute to 198.51.100.143 (198.51.100.143), 30 hops max, 60 byte packets
 1  192.168.20.1 (192.168.20.1)  19.561 ms  *  *
 2  192.168.20.1 (192.168.20.1)  241.525 ms  302.570 ms  329.466 ms
 3  * 203.0.113.2 (203.0.113.2)  130.778 ms  *
 4  * * 198.51.100.2 (198.51.100.2)  135.711 ms
 5  * * *
 6  203.0.113.5 (203.0.113.5)  223.106 ms  552.471 ms  185.721 ms
 7  198.51.100.143 (198.51.100.143)  1107.954 ms  979.324 ms  979.498 ms
```

```
vyos@cl-3344:~$
vyos@vyos-fw:~$ show int
Codes: S - State, L - Link, u - Up, D - Down, A - Admin Down
Interface      IP Address      MAC              VRF      MTU  S/L  Description
-----
eth0            -              50:01:00:06:00:00 default  1500 u/u   trunk-to-bl-1
eth0.100        203.0.113.2/30  50:01:00:06:00:00 TENANT-1  1500 u/u   TENANT-1 transit to bl-1
eth0.130        203.0.113.6/30  50:01:00:06:00:00 IDRAC    1500 u/u   IDRAC transit to bl-1
eth1            198.51.100.1/30 50:01:00:06:00:01 default  1500 u/u   ISP
eth2            -              50:01:00:06:00:02 default  1500 u/u
```

```
bl-1#
bl-1#show ip route vrf IDRAC

VRF: IDRAC
Codes: C - connected, S - static, K - kernel,
       O - OSPF, IA - OSPF inter area, E1 - OSPF external type 1,
       E2 - OSPF external type 2, N1 - OSPF NSSA external type 1,
       N2 - OSPF NSSA external type2, B - Other BGP Routes,
       B I - iBGP, B E - eBGP, R - RIP, I L1 - IS-IS level 1,
       I L2 - IS-IS level 2, O3 - OSPFv3, A B - BGP Aggregate,
       A O - OSPF Summary, NG - Nexthop Group Static Route,
       V - VXLAN Control Service, M - Martian,
       DH - DHCP client installed default route,
       DP - Dynamic Policy Route, L - VRF Leaked,
       G - gRIBI, RC - Route Cache Route
```

```
Gateway of last resort:
B E      0.0.0.0/0 [200/0] via 203.0.113.6, Vlan130
```

```
C      198.51.100.128/25 is directly connected, Vlan30
C      203.0.113.4/30 is directly connected, Vlan130
```

```
bl-1#show arp vrf IDRAC
```

	Age (sec)	Hardware Addr	Interface
.130	3:08:14	0050.7966.6812	Vlan30, Ethernet4
.143	2:58:37	0050.7966.6809	Vlan30, not learned
6	0:00:00	5001.0006.0000	Vlan130, Ethernet3

DC2 / POD

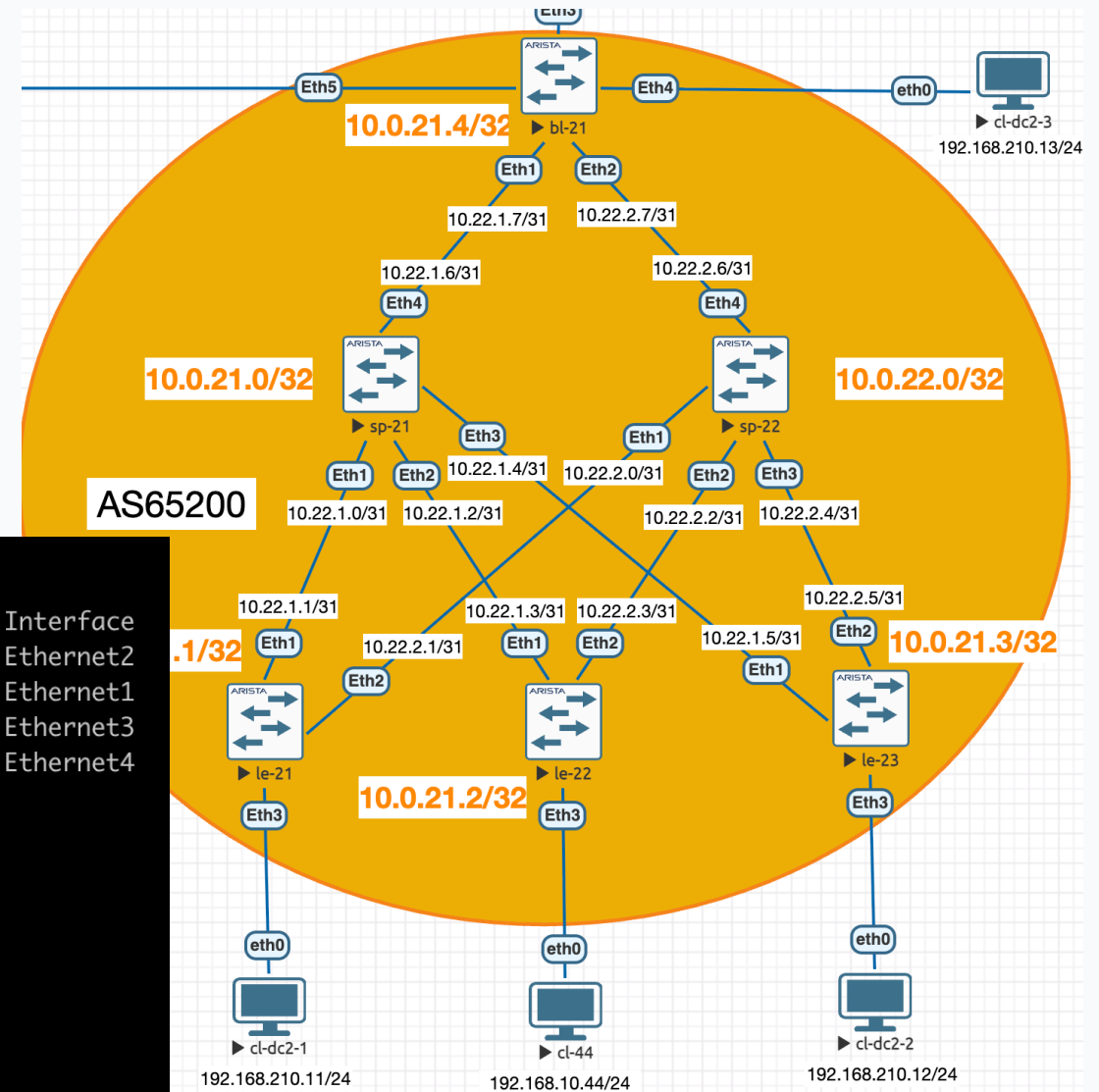
- DC2 построен как отдельный fabric
- Underlay DC2 — OSPF area 0
- Overlay DC2 — iBGP EVPN в AS 65200
- sp-21 и sp-22 работают как route-reflector
- TENANT-DC2: VLAN 210, L2VNI 10210, L3VNI 50200

```
sp-21#
sp-21#show ip ospf neighbor
Neighbor ID      Instance VRF      Pri State      Dead Time   Address      Interface
10.0.21.2        1       default  0  FULL        00:00:33    10.22.1.3    Ethernet2
10.0.21.1        1       default  0  FULL        00:00:30    10.22.1.1    Ethernet1
10.0.21.3        1       default  0  FULL        00:00:38    10.22.1.5    Ethernet3
10.0.21.4        1       default  0  FULL        00:00:29    10.22.1.7    Ethernet4

sp-21#show bgp evpn summary
BGP summary information for VRF default
Router identifier 10.0.21.0, local AS number 65200
Neighbor Status Codes: m - Under maintenance

Neighbor  V AS      MsgRcvd  MsgSent  InQ  OutQ  Up/Down  State  PfxRcd  PfxAcc
10.0.21.1 4 65200    14883    15793    0    0  07:57:34 Estab   4       4
10.0.21.2 4 65200    14822    15838    0    0  07:57:23 Estab   4       4
10.0.21.3 4 65200    14895    15972    0    0  07:57:58 Estab   4       4
10.0.21.4 4 65200    15006    14420    0    0  07:58:13 Estab  46      46

sp-21#
```



DCI между DC1 и DC2

- DCI построен между bl-1 и bl-21
- Underlay DCI — eBGP IPv4 по 10.255.0.0/31
- Overlay DCI — eBGP EVPN по loopback
- Первый сценарий DCI — L2 stretch VLAN 10
- cl-44 в DC2 использует gateway DC1 и выходит в интернет через DC1

```
bl-1#show ip bgp summary
BGP summary information for VRF default
Router identifier 10.0.1.4, local AS number 65104
Neighbor Status Codes: m - Under maintenance
Neighbor  V AS      MsgRcvd  MsgSent  InQ OutQ  Up/Down State  PfxRcd PfxAcc
10.2.1.6   4 65001    13952    13986    0   0 08:02:29 Estab   5      5
10.2.2.6   4 65001    13940    13985    0   0 08:02:43 Estab   5      5
10.255.0.1 4 65200    13740    13852    0   0 08:02:30 Estab   5      5
bl-1#show bgp evpn summary
BGP summary information for VRF default
Router identifier 10.0.1.4, local AS number 65104
Neighbor Status Codes: m - Under maintenance
Neighbor  V AS      MsgRcvd  MsgSent  InQ OutQ  Up/Down State  PfxRcd PfxAcc
10.0.1.0   4 65001    14745    14743    0   0 09:51:21 Estab  33     33
10.0.2.0   4 65001    14722    14724    0   0 09:51:25 Estab  33     33
10.0.21.4  4 65200    13755    14564    0   0 08:03:03 Estab  16     16
bl-1#
```

```
cl-44> show arp
```

```
00:00:be:ef:ca:fe 192.168.10.1 expires in 118 seconds
```

```
cl-44> ping 192.168.10.12
```

```
84 bytes from 192.168.10.12 icmp_seq=1 ttl=64 time=369.604 ms
84 bytes from 192.168.10.12 icmp_seq=2 ttl=64 time=63.569 ms
84 bytes from 192.168.10.12 icmp_seq=3 ttl=64 time=71.114 ms
84 bytes from 192.168.10.12 icmp_seq=4 ttl=64 time=95.187 ms
84 bytes from 192.168.10.12 icmp_seq=5 ttl=64 time=84.048 ms
```

```
cl-44> ping 8.8.8.8
```

```
84 bytes from 8.8.8.8 icmp_seq=1 ttl=62 time=115.839 ms
84 bytes from 8.8.8.8 icmp_seq=2 ttl=62 time=182.290 ms
84 bytes from 8.8.8.8 icmp_seq=3 ttl=62 time=282.816 ms
84 bytes from 8.8.8.8 icmp_seq=4 ttl=62 time=184.822 ms
84 bytes from 8.8.8.8 icmp_seq=5 ttl=62 time=130.851 ms
```

```
cl-44> █
```

Проверки и демонстрация

- Control-plane: BGP/OSPF/EVPN соседства
- Routing: VRF routes, default route через border leaf
- Data-plane: ping и trace между клиентами
- Application: curl HTTP и iperf3 до 8.8.8.8
- DCI: VLAN 10 stretch и интернет для cl-44

```
vyos@cl-3344:~$  
vyos@cl-3344:~$  
vyos@cl-3344:~$ curl 8.8.8.8  
<h1>Internet</h1>  
<p>Hello from simulated Internet!</p>  
vyos@cl-3344:~$
```

```
cl-33 (telnet)
84 bytes from 8.8.8.8 icmp_seq=24384 ttl=61 time=68.160 ms
84 bytes from 8.8.8.8 icmp_seq=24385 ttl=61 time=66.452 ms
84 bytes from 8.8.8.8 icmp_seq=24386 ttl=61 time=128.606 ms
84 bytes from 8.8.8.8 icmp_seq=24387 ttl=61 time=67.770 ms
84 bytes from 8.8.8.8 icmp_seq=24388 ttl=61 time=98.061 ms
84 bytes from 8.8.8.8 icmp_seq=24389 ttl=61 time=172.490 ms
84 bytes from 8.8.8.8 icmp_seq=24390 ttl=61 time=744.248 ms
84 bytes from 8.8.8.8 icmp_seq=24391 ttl=61 time=223.364 ms
84 bytes from 8.8.8.8 icmp_seq=24392 ttl=61 time=260.215 ms
84 bytes from 8.8.8.8 icmp_seq=24393 ttl=61 time=575.389 ms
84 bytes from 8.8.8.8 icmp_seq=24394 ttl=61 time=654.681 ms
84 bytes from 8.8.8.8 icmp_seq=24395 ttl=61 time=938.698 ms
84 bytes from 8.8.8.8 icmp_seq=24396 ttl=61 time=677.629 ms
8.8.8.8 icmp_seq=24397 timeout
8.8.8.8 icmp_seq=24398 timeout
8.8.8.8 icmp_seq=24399 timeout
8.8.8.8 icmp_seq=24400 timeout
8.8.8.8 icmp_seq=24401 timeout
8.8.8.8 icmp_seq=24402 timeout
8.8.8.8 icmp_seq=24403 timeout
84 bytes from 8.8.8.8 icmp_seq=24404 ttl=61 time=60.541 ms
84 bytes from 8.8.8.8 icmp_seq=24405 ttl=61 time=57.855 ms
84 bytes from 8.8.8.8 icmp_seq=24406 ttl=61 time=54.114 ms
84 bytes from 8.8.8.8 icmp_seq=24407 ttl=61 time=51.309 ms
84 bytes from 8.8.8.8 icmp_seq=24408 ttl=61 time=66.059 ms
84 bytes from 8.8.8.8 icmp_seq=24409 ttl=61 time=77.192 ms
84 bytes from 8.8.8.8 icmp_seq=24410 ttl=61 time=67.896 ms
84 bytes from 8.8.8.8 icmp_seq=24411 ttl=61 time=63.702 ms
84 bytes from 8.8.8.8 icmp_seq=24412 ttl=61 time=154.944 ms
[]

cl-11 (telnet)
84 bytes from 198.51.100.129 icmp_seq=82 ttl=59 time=632.545 ms
84 bytes from 198.51.100.129 icmp_seq=83 ttl=59 time=66.194 ms
84 bytes from 198.51.100.129 icmp_seq=84 ttl=59 time=61.673 ms
84 bytes from 198.51.100.129 icmp_seq=85 ttl=59 time=69.818 ms
84 bytes from 198.51.100.129 icmp_seq=86 ttl=59 time=66.497 ms
84 bytes from 198.51.100.129 icmp_seq=87 ttl=59 time=215.604 ms
84 bytes from 198.51.100.129 icmp_seq=88 ttl=59 time=788.398 ms
84 bytes from 198.51.100.129 icmp_seq=89 ttl=59 time=285.691 ms
84 bytes from 198.51.100.129 icmp_seq=90 ttl=59 time=157.267 ms
84 bytes from 198.51.100.129 icmp_seq=91 ttl=59 time=567.106 ms
84 bytes from 198.51.100.129 icmp_seq=92 ttl=59 time=577.083 ms
84 bytes from 198.51.100.129 icmp_seq=93 ttl=59 time=916.812 ms
84 bytes from 198.51.100.129 icmp_seq=94 ttl=59 time=813.695 ms
198.51.100.129 icmp_seq=95 timeout
198.51.100.129 icmp_seq=96 timeout
198.51.100.129 icmp_seq=97 timeout
84 bytes from 198.51.100.129 icmp_seq=98 ttl=59 time=901.640 ms
198.51.100.129 icmp_seq=99 timeout
198.51.100.129 icmp_seq=100 timeout
198.51.100.129 icmp_seq=101 timeout
84 bytes from 198.51.100.129 icmp_seq=102 ttl=59 time=134.215 ms
84 bytes from 198.51.100.129 icmp_seq=103 ttl=59 time=142.566 ms
84 bytes from 198.51.100.129 icmp_seq=104 ttl=59 time=64.893 ms
84 bytes from 198.51.100.129 icmp_seq=105 ttl=59 time=81.139 ms
84 bytes from 198.51.100.129 icmp_seq=106 ttl=59 time=69.219 ms
84 bytes from 198.51.100.129 icmp_seq=107 ttl=59 time=95.643 ms
84 bytes from 198.51.100.129 icmp_seq=108 ttl=59 time=64.526 ms
84 bytes from 198.51.100.129 icmp_seq=109 ttl=59 time=68.643 ms
84 bytes from 198.51.100.129 icmp_seq=110 ttl=59 time=56.411 ms
[]

cl-11drac (telnet)
84 bytes from 1.1.1.1 icmp_seq=25388 ttl=62 time=137.668 ms
84 bytes from 1.1.1.1 icmp_seq=25389 ttl=62 time=133.912 ms
84 bytes from 1.1.1.1 icmp_seq=25390 ttl=62 time=128.352 ms
84 bytes from 1.1.1.1 icmp_seq=25391 ttl=62 time=113.496 ms
84 bytes from 1.1.1.1 icmp_seq=25392 ttl=62 time=152.156 ms
84 bytes from 1.1.1.1 icmp_seq=25393 ttl=62 time=110.990 ms
84 bytes from 1.1.1.1 icmp_seq=25394 ttl=62 time=141.883 ms
84 bytes from 1.1.1.1 icmp_seq=25395 ttl=62 time=116.286 ms
84 bytes from 1.1.1.1 icmp_seq=25396 ttl=62 time=162.345 ms
84 bytes from 1.1.1.1 icmp_seq=25397 ttl=62 time=154.583 ms
84 bytes from 1.1.1.1 icmp_seq=25398 ttl=62 time=115.095 ms
84 bytes from 1.1.1.1 icmp_seq=25399 ttl=62 time=112.159 ms
84 bytes from 1.1.1.1 icmp_seq=25400 ttl=62 time=177.777 ms
84 bytes from 1.1.1.1 icmp_seq=25401 ttl=62 time=192.108 ms
84 bytes from 1.1.1.1 icmp_seq=25402 ttl=62 time=142.658 ms
84 bytes from 1.1.1.1 icmp_seq=25403 ttl=62 time=308.019 ms
84 bytes from 1.1.1.1 icmp_seq=25404 ttl=62 time=344.568 ms
1.1.1.1 icmp_seq=25405 timeout
1.1.1.1 icmp_seq=25406 timeout
1.1.1.1 icmp_seq=25407 timeout
84 bytes from 1.1.1.1 icmp_seq=25408 ttl=62 time=246.046 ms
84 bytes from 1.1.1.1 icmp_seq=25409 ttl=62 time=38.082 ms
84 bytes from 1.1.1.1 icmp_seq=25410 ttl=62 time=26.683 ms
84 bytes from 1.1.1.1 icmp_seq=25411 ttl=62 time=47.289 ms
84 bytes from 1.1.1.1 icmp_seq=25412 ttl=62 time=29.509 ms
84 bytes from 1.1.1.1 icmp_seq=25413 ttl=62 time=28.452 ms
84 bytes from 1.1.1.1 icmp_seq=25414 ttl=62 time=112.226 ms
84 bytes from 1.1.1.1 icmp_seq=25415 ttl=62 time=29.442 ms
84 bytes from 1.1.1.1 icmp_seq=25416 ttl=62 time=48.958 ms
[]

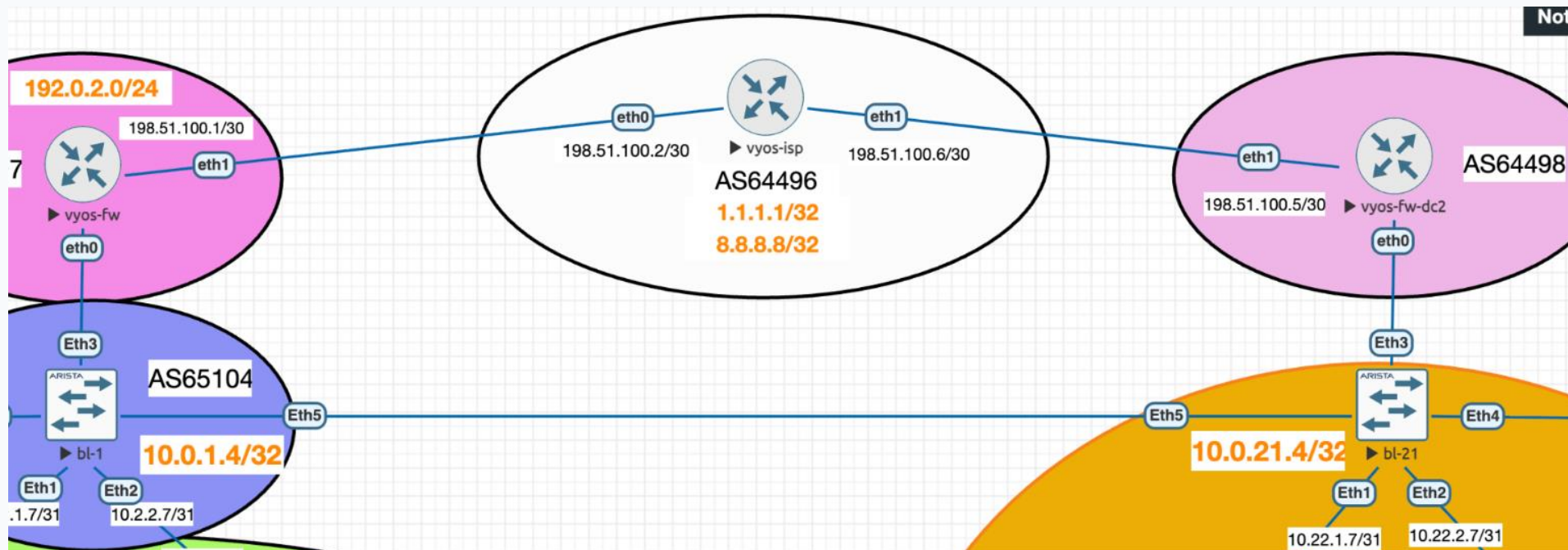
cl-11dc2-1 (telnet)
84 bytes from 192.168.210.12 icmp_seq=32553 ttl=64 time=35.687 ms
84 bytes from 192.168.210.12 icmp_seq=32554 ttl=64 time=51.530 ms
84 bytes from 192.168.210.12 icmp_seq=32555 ttl=64 time=49.782 ms
84 bytes from 192.168.210.12 icmp_seq=32556 ttl=64 time=48.373 ms
84 bytes from 192.168.210.12 icmp_seq=32557 ttl=64 time=49.328 ms
84 bytes from 192.168.210.12 icmp_seq=32558 ttl=64 time=37.049 ms
84 bytes from 192.168.210.12 icmp_seq=32559 ttl=64 time=52.687 ms
84 bytes from 192.168.210.12 icmp_seq=32560 ttl=64 time=42.245 ms
84 bytes from 192.168.210.12 icmp_seq=32561 ttl=64 time=33.956 ms
84 bytes from 192.168.210.12 icmp_seq=32562 ttl=64 time=36.772 ms
84 bytes from 192.168.210.12 icmp_seq=32563 ttl=64 time=33.531 ms
84 bytes from 192.168.210.12 icmp_seq=32564 ttl=64 time=32.404 ms
84 bytes from 192.168.210.12 icmp_seq=32565 ttl=64 time=90.677 ms
84 bytes from 192.168.210.12 icmp_seq=32566 ttl=64 time=43.489 ms
84 bytes from 192.168.210.12 icmp_seq=32567 ttl=64 time=47.800 ms
84 bytes from 192.168.210.12 icmp_seq=32568 ttl=64 time=46.297 ms
84 bytes from 192.168.210.12 icmp_seq=32569 ttl=64 time=43.786 ms
84 bytes from 192.168.210.12 icmp_seq=32570 ttl=64 time=47.817 ms
84 bytes from 192.168.210.12 icmp_seq=32571 ttl=64 time=67.107 ms
84 bytes from 192.168.210.12 icmp_seq=32572 ttl=64 time=47.716 ms
84 bytes from 192.168.210.12 icmp_seq=32573 ttl=64 time=41.074 ms
84 bytes from 192.168.210.12 icmp_seq=32574 ttl=64 time=59.323 ms
84 bytes from 192.168.210.12 icmp_seq=32575 ttl=64 time=103.477 ms
84 bytes from 192.168.210.12 icmp_seq=32576 ttl=64 time=44.951 ms
84 bytes from 192.168.210.12 icmp_seq=32577 ttl=64 time=35.219 ms
84 bytes from 192.168.210.12 icmp_seq=32578 ttl=64 time=42.750 ms
84 bytes from 192.168.210.12 icmp_seq=32579 ttl=64 time=37.741 ms
84 bytes from 192.168.210.12 icmp_seq=32580 ttl=64 time=55.383 ms
84 bytes from 192.168.210.12 icmp_seq=32581 ttl=64 time=40.763 ms
[]

cl-3344 (telnet)
es
[ 11] 19.00-20.00 sec 128 KBytes 1.05 Mbits/sec 7 9.90 KB
ytes
[SUM] 19.00-20.00 sec 128 KBytes 1.05 Mbits/sec 25
- - - - -
[ ID] Interval Transfer Bitrate Retr
[ 5] 0.00-20.00 sec 225 KBytes 92.1 Kbits/sec 39
sender
[ 5] 0.00-21.35 sec 99.0 KBytes 38.0 Kbits/sec
receiver
[ 7] 0.00-20.00 sec 384 KBytes 157 Kbits/sec 56
sender
[ 7] 0.00-21.35 sec 180 KBytes 68.9 Kbits/sec
receiver
[ 9] 0.00-20.00 sec 768 KBytes 315 Kbits/sec 58
sender
[ 9] 0.00-21.35 sec 397 KBytes 152 Kbits/sec
receiver
[ 11] 0.00-20.00 sec 384 KBytes 157 Kbits/sec 67
sender
[ 11] 0.00-21.35 sec 137 KBytes 52.6 Kbits/sec
receiver
[SUM] 0.00-20.00 sec 1.72 MBytes 721 Kbits/sec 220
sender
[SUM] 0.00-21.35 sec 813 KBytes 312 Kbits/sec
receiver
iperf Done.
vyos@cl-3344:~$

cl-44 (telnet)
84 bytes from 192.168.10.12 icmp_seq=70 ttl=64 time=634.555 ms
84 bytes from 192.168.10.12 icmp_seq=71 ttl=64 time=79.654 ms
84 bytes from 192.168.10.12 icmp_seq=72 ttl=64 time=74.788 ms
84 bytes from 192.168.10.12 icmp_seq=73 ttl=64 time=78.893 ms
84 bytes from 192.168.10.12 icmp_seq=74 ttl=64 time=134.536 ms
84 bytes from 192.168.10.12 icmp_seq=75 ttl=64 time=120.221 ms
84 bytes from 192.168.10.12 icmp_seq=76 ttl=64 time=700.875 ms
84 bytes from 192.168.10.12 icmp_seq=77 ttl=64 time=392.996 ms
84 bytes from 192.168.10.12 icmp_seq=78 ttl=64 time=164.651 ms
84 bytes from 192.168.10.12 icmp_seq=79 ttl=64 time=639.641 ms
84 bytes from 192.168.10.12 icmp_seq=80 ttl=64 time=543.332 ms
84 bytes from 192.168.10.12 icmp_seq=81 ttl=64 time=693.567 ms
84 bytes from 192.168.10.12 icmp_seq=82 ttl=64 time=885.533 ms
192.168.10.12 icmp_seq=83 timeout
192.168.10.12 icmp_seq=84 timeout
192.168.10.12 icmp_seq=85 timeout
192.168.10.12 icmp_seq=86 timeout
192.168.10.12 icmp_seq=87 timeout
192.168.10.12 icmp_seq=88 timeout
192.168.10.12 icmp_seq=89 timeout
84 bytes from 192.168.10.12 icmp_seq=90 ttl=64 time=151.437 ms
84 bytes from 192.168.10.12 icmp_seq=91 ttl=64 time=90.555 ms
84 bytes from 192.168.10.12 icmp_seq=92 ttl=64 time=75.525 ms
84 bytes from 192.168.10.12 icmp_seq=93 ttl=64 time=85.151 ms
84 bytes from 192.168.10.12 icmp_seq=94 ttl=64 time=82.535 ms
84 bytes from 192.168.10.12 icmp_seq=95 ttl=64 time=87.007 ms
84 bytes from 192.168.10.12 icmp_seq=96 ttl=64 time=80.763 ms
84 bytes from 192.168.10.12 icmp_seq=97 ttl=64 time=88.586 ms
84 bytes from 192.168.10.12 icmp_seq=98 ttl=64 time=138.268 ms
[]
```

Ограничения и дальнейшее развитие

- В каждом DC используется один border leaf
- DCI построен одним линком
- IDRAC gateway 198.51.100.129/25 расположен только на bl-1
- В DC2 нет отдельной public /24; NAT выполняется в линковочную public-подсеть
- Развитие: второй border leaf, резервный DCI, HA edge, расширенные failure-тесты



Вывод

- Собран рабочий VXLAN EVPN fabric
- Реализованы L2VNI, L3VNI и Anycast Gateway
- Добавлен внешний edge с VyOS, NAT, firewall и BGP-фильтрацией
- Проверены MLAG, EVPN multihoming, IDRAC VRF, DC2 и DCI
- Архитектура масштабируема и допускает дальнейшее развитие

Слой	Технология	Назначение
Underlay	eBGP IPv4	IP-достижимость loopback/VTEP между leaf и spine
DC2 Underlay	OSPF	IP-достижимость loopback/VTEP внутри второго POD
Overlay control plane	MP-BGP EVPN	Распространение MAC/IP, IMET и IP-prefix маршрутов
DC2 Overlay	iBGP EVPN RR	Spine DC2 работают как route-reflector для leaf DC2
DCI Underlay	eBGP IPv4	IP-достижимость loopback/VTEP между border leaf DC1 и DC2
DCI Overlay	eBGP EVPN	Межплощадочный EVPN control-plane между b1-1 и b1-21
Overlay data plane	VXLAN	Инкапсуляция клиентского L2/L3-трафика
BGP separation	Underlay/Overlay peer-groups	IPv4-сессии по p2p-адресам, EVPN-сессии по Loopback0
Tenant routing	VRF TENANT-1	Изоляция клиентских маршрутов
OOB routing	VRF IDRAC	Отдельный публичный OOB/iDRAC-сегмент
L2 service	L2VNI	Растягивание VLAN поверх fabric
L3 service	L3VNI	Маршрутизация между VLAN внутри tenant VRF
Gateway	Anycast Gateway	Единый шлюз VLAN на всех leaf
Edge	VyOS NAT	Выход tenant-сетей во внешний сегмент
Filtering	VyOS firewall, prefix-list, route-map	Ограничение входящего доступа и коррекция BGP-анонсов
Custom anycast MAC	00:00:be:ef:ca:fe , 00:00:be:ef:dc:02	Раздельные virtual-router MAC для DC1 и DC2
Client HA	MLAG + LACP	Dual-homed подключение c1-1122 к паре le-1 / le-2
Client HA	EVPN multihoming / ESI-LAG	Dual-homed подключение c1-3344 к le-3 / le-4 без MLAG
Failure handling	Link Tracking	Отслеживание uplink/downlink на leaf для отказоустойчивых клиентских портов